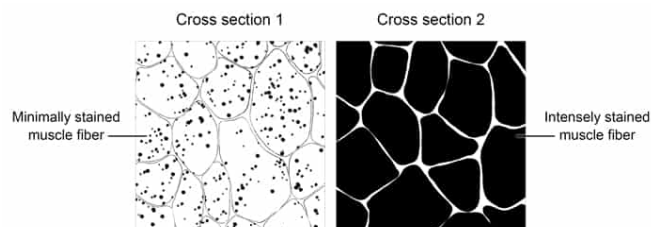


In an assay, researchers assessed activity levels of an oxidative enzyme present on the inner mitochondrial membrane of skeletal muscle fibers. Enzyme activity was visualized by reacting the product of the enzymatic reaction with a dark-colored salt, which resulted in dark-colored staining of the muscle fibers. Assay results in two different microscopic cross sections of muscle fibers are shown below.



Given these results, muscle fibers in cross section 1 most likely:

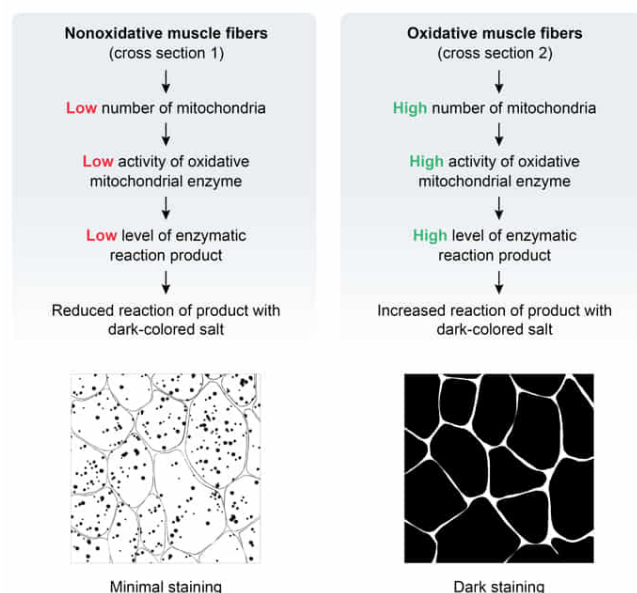
- ☐ A. utilized more oxygen than muscle fibers in cross section 2. [5%]
- ☐ B. relied on anaerobic respiration less than muscle fibers in cross section 2. [12%]
- ☒ C. contained fewer oxygen-storage proteins than muscle fibers in cross section 2. [76%]
- ☐ D. had a more extensive network of surrounding blood vessels than muscle fibers in cross section 2. [5%]

Correct

75% answered correctly

Explanation:

Activity of mitochondrial enzyme in nonoxidative & oxidative muscle fibers



Skeletal muscle fibers are classified according to how rapidly they contract, how they procure the ATP necessary for contraction, and how resistant they are to fatigue with repeated stimulation. **Muscle fiber types** include the following:

- **Slow-twitch** (type 1) **oxidative fibers** contract at a slow rate due to the lesser frequency at which the myosin heads **hydrolyze ATP** during each contractile cycle. These fibers produce ATP via **aerobic respiration**. Therefore, they contain *many* mitochondria that use oxygen in the electron transport chain to produce *large* amounts of ATP (energy), making these fibers *more* fatigue resistant. Type 1 fibers receive oxygen from their *extensive* capillary networks and *high* concentrations of **myoglobin**.
- **Fast-twitch fibers** achieve more rapid contraction rates (complete more contractile cycles) than slow-twitch fibers because they can hydrolyze ATP much faster. These fibers are classified as follows:
 - **Fast glycolytic** (type 2X) **fibers** rely *primarily* on **nonoxidative (anaerobic) glycolysis** for ATP production. Accordingly, these fibers fatigue more easily due to the acidosis caused by H^+ accumulation from **anaerobic ATP hydrolysis**. Because type 2X fibers generally contain *very few* mitochondria and require much *less* oxygen than oxidative fibers, they exhibit *low* myoglobin concentration and *less* extensive capillary networks.
 - **Fast oxidative–glycolytic** (type 2A) **fibers** use *both* oxidative (ie, aerobic respiration) and nonoxidative processes (ie, anaerobic glycolysis) to generate ATP, making them moderately susceptible to fatigue. Like type 1 fibers, these fibers also receive oxygen from abundant myoglobin reserves and extensive capillary networks.

In this scenario, researchers assessed the activity of an **oxidative enzyme** on the mitochondrial membrane of skeletal muscle fibers. Reacting a dark-colored salt with the enzymatic reaction product stained fibers with high enzymatic activity (and product levels) a dark color. Therefore, darkly stained fibers in *cross section 2* indicate high enzyme activity and the presence of *many* mitochondria, making them *mostly oxidative*. In contrast, minimally stained fibers in cross section 1 indicate low enzymatic activity (ie, fewer mitochondria), signifying that these fibers are primarily **nonoxidative**.

Consequently, nonoxidative fibers (cross section 1) likely utilized *less* oxygen (**Choice A**), relied *more* on anaerobic respiration (**Choice B**), had *less* extensive networks of surrounding blood vessels (**Choice D**), and had **fewer oxygen-storage proteins** (myoglobin) than oxidative fibers (cross section 2).

Educational objective:

Oxidative muscle fibers use aerobic respiration for ATP synthesis, requiring large amounts of oxygen supplied by high levels of myoglobin and extensive capillary networks. In contrast, nonoxidative fibers generate ATP through anaerobic glycolysis and require less oxygen than oxidative fibers.

Time spent:0 sec

Subject:

Biology

QID:402463

System:

Musculoskeletal System